NETWORK VERSION CONTROL SYSTEM (PES-VCS)

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SRN: PES1UG24CS054.
Topic: Version Control System Implementation.

1A.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$
```

2A.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ echo "hello" > file1.txt
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes add file1.txt
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt

Unstaged changes:
  modified:   file1.txt

Untracked files:
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  commit.h
  untracked:  .gitignore
  untracked:  index.h
  untracked:  tree.h
  untracked:  pes.c
  untracked:  Makefile
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  pes.h
  untracked:  commit.c
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  object.c

kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ cat .pex/index
cat: .pex/index: No such file or directory
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 6 file1.txt
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$
```

3A.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ make pes
make: 'pes' is up to date.
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ echo "hello" > file1.txt
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ echo "world" > file2.txt
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes add file1.txt file2.txt
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  commit.h
  untracked:  .gitignore
  untracked:  index.h
  untracked:  tree.h
  untracked:  pes.c
  untracked:  Makefile
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  pes.h
  untracked:  commit.c
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  object.c

kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$
```

3B.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776693183 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776693193 6 file2.txt
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$
```

4A.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"
Committed: 539319efa6e5... Initial commit
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"
Committed: 8e22458ecdd5... Add world
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ echo "Bye" > bye.txt
./pes add bye.txt
./pes commit -m "Add bye file"
Committed: d6e3b6148836... Add bye file
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ ./pes log
commit d6e3b6148836e4573df94e97de993e4148be518ca5175f71809b389639214f65
Author: PES User <pes@localhost>
Date:   1776693751

    Add bye file

commit 8e22458ecdd5ec7a08445c3f379d759dac4ae5bad93054e859e2f4682ebfce17
Author: PES User <pes@localhost>
Date:   1776693742

    Add worldile

commit 539319efa6e5ff92252a79176924b7e705c62d25c7ce192120b112e7c9fe2f5c
Author: PES User <pes@localhost>
Date:   1776693734

    Initial committial commiQ
```

4B.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/53/9319efa6e5ff92252a79176924b7e705c62d25c7ce192120b112e7c9fe2f5c
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/8e/22458ecdd5ec7a08445c3f379d759dac4ae5bad93054e859e2f4682ebfce17
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d6/e3b6148836e4573df94e97de993e4148be518ca5175f71809b389639214f65
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/f3/97707e14d1cabf1c29a72d833bffb94d4a474ecc48789d6496742ae6541034
.pes/refs/heads/main
```

4C.

```
kattimaniamrutha@LAPTOP-1IMGKELC:~/PES1UG24CS054-pes-vcs$ cat .pes/HEAD
cat .pes/refs/heads/main
ref: refs/heads/main
d6e3b6148836e4573df94e97de993e4148be518ca5175f71809b389639214f65
```

FINAL:

Phase 5 & 6 – Answers

Q5.1

To implement pes checkout <branch>:

- **Update .pes/HEAD to point to the selected branch**
- **Read the commit hash from .pes/refs/heads/<branch>**
- **Load the commit and get its tree**
- **Recreate the working directory from the tree contents**
- **Update the index to match the checked-out state**

Complexity:

- **Existing files may be overwritten**
- **Uncommitted changes must be handled carefully**
- **Directory structure must be rebuilt correctly**

Q5.2

To detect a dirty working directory:

- **Compare index entries with working directory files**
- **For each file:**
    - **Check modification time and size**
    - **If different, recompute hash and compare with index**

If hashes differ, the file is modified

Also:

- **Compare with the target branch's tree**
- **If the same file differs there as well, it is a conflict**

In such cases, checkout should be refused

Q5.3

Detached HEAD means:

- **HEAD points directly to a commit instead of a branch**

If commits are made:

- **New commits are created**
- **No branch points to them**

Recovery:

- **Create a branch from that commit using its hash**

Q6.1

Garbage collection algorithm:

1. **Start from all branch heads**
2. **Traverse all commits, trees, and blobs**
3. **Mark all reachable objects**
4. **Delete unmarked objects from .pes/objects/**

Data structure:

- **Use a hash set for efficient lookup**

Estimate:

- **Around 100,000 commits**
- **Total objects visited roughly 200,000 to 300,000**

Q6.2

Garbage collection is dangerous during commit because:

- **Objects are written before references are updated**
- **If GC runs at that time, it may treat new objects as unreachable**
- **Those objects may get deleted**

This can break the repository

Git avoids this by:

- **Using locks**
- **Writing objects before updating references**
- **Using atomic operations like temp file and rename**
- **Avoiding deletion of recent objects**